CS3235 — LLM / Multimodal AI Projects

Mini-Project 3

PatraAI

Indian Document Understanding Tool

Final Project Report

Extracting fields, summarising, translating and answering questions about Indian government documents

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Abstract

PatraAI is a web application that helps people understand Indian government documents such as notices, circulars, scholarship letters, certificates and application forms. The user uploads a PDF, Word file or image. PatraAI extracts the key details with the page each came from, writes a short summary, translates the results into Kannada, Hindi, Tamil, Telugu or Malayalam with AI4Bharat IndicTrans2, and answers questions asked in English or any of these five languages.

BharatGen Patram, the vision-language model named in the project sheet, needs more memory than the project laptop has, so a smaller multimodal model (qwen3.5:4b, run with Ollama) does the reading and answering. The application can switch to Patram through one setting when a GPU server is available. Everything runs on a laptop without a GPU.

On 45 hand-checked test cases from 18 real public documents, the first evaluation gave 64.4% field-extraction accuracy and 66.7% question-answering accuracy. The failure analysis showed that most errors came from scanned Kannada and Malayalam pages that OCR read badly. After adding Tesseract OCR with Indian-language packs and fixing the date check, accuracy rose to 75.6% for fields and 77.8% for questions.

Table 1**:** Key results

What was measured	Result
Test cases	45 cases over 18 real public documents in 6 languages
Field-extraction accuracy	75.6% (34 of 45); first run 64.4%
Question-answering (QA) accuracy	77.8% (35 of 45); first run 66.7%
Answer cited the correct page	77.8%
Automated software tests	98, all passing
Speed-up from optimisation	analysis 23% less time, translation 50% less time

Keywords: document understanding, OCR, multimodal LLM, retrieval-augmented generation, IndicTrans2, Indian languages

Table of Contents

Abstract ii
Table of Contents iii
1. Problem Definition and Relevance 1
1.1 Why this is hard 1
2. Model and Dataset Selection 1
2.1 BharatGen Patram 2
2.2 AI4Bharat IndicTrans2 2
2.3 Why qwen3.5:4b 2
2.4 Evaluation dataset 2
3. Working End-to-End Application 2
3.1 Features 4
4. GenAI, RAG and Agentic Integration 4
4.1 Structured extraction 4
4.2 Question answering with retrieval (RAG) 4
4.3 Agentic loop for the answer language 4
4.4 Checks against hallucination 4
5. Quantitative Testing and Error Analysis 5
5.1 Method 5
5.2 Results 5
5.3 Failure analysis 6
5.4 Speed 7
5.5 Automated tests 8
6. Demo, Documentation and Repository 8
6.1 How to run 8
6.2 Demo script 8
7. Limitations and Future Work 9
8. Conclusion 9
References 9
Appendix A. Test Cases and Results 10

1. Problem Definition and Relevance

Government notices, circulars, scholarship letters, certificates and application forms decide who is eligible for something, which documents to submit and by when. In India they are issued in English and in regional languages, are often scanned, and can run to many pages. A reader who knows only one of these languages may not be able to find the key details.

PatraAI is a general tool for any such document. It shows the key details with the page they came from, a short summary, translations and answers to questions. When the document does not contain an answer, it says so instead of guessing.

1.1 Why this is hard

- **Many scripts and old fonts.** Six scripts are supported, and many Kannada PDFs use old fonts whose text cannot be copied correctly, so those pages must be read with OCR.
- **Scanned pages.** Pages that are only pictures need OCR before anything else can happen.
- **Small model, no GPU.** On a laptop CPU the model writes only about 9 tokens per second, so prompts must be short and its output must be checked for invented numbers.
- **Cross-language questions.** A question in Kannada about an English document must be answered in Kannada from English text.

2. Model and Dataset Selection

The project sheet specifies BharatGen Patram and AI4Bharat IndicTrans2, and asks to check each model's availability, licence, languages and GPU needs before using it.

Table 2: Models and their roles

Task	Model	Used?
Read documents, extract fields, summarise, answer questions	BharatGen Patram 7B (specified)	No: needs a GPU server
Same tasks (actually used)	qwen3.5:4b via Ollama, on the laptop CPU	Yes
Translate English into 5 Indian languages	AI4Bharat IndicTrans2 (200M), on the laptop CPU	Yes
Read scanned pages (OCR)	Tesseract 5.4 with Indian-language packs; the vision model as fallback	Yes (after fixes)
Backup if a model reply fails	gemma4:e4b	Only as a retry

2.1 BharatGen Patram

Patram-7B needs about 15.3 GB of memory for its weights alone, but the laptop has 15.2 GB of RAM and no GPU. There is no hosted service or smaller (quantised) version, and Ollama cannot run its custom architecture, so it could not run here. PatraAI still supports it: one setting (HF_CHAT_COMPLETIONS_URL) sends all requests to a Patram server, and the application always shows which model actually answered.

2.2 AI4Bharat IndicTrans2

IndicTrans2 translates the summary, the extracted details and the answers from English into Kannada, Hindi, Tamil, Telugu and Malayalam on the CPU. Numbers, dates and codes are protected so translation never changes them.

2.3 Why qwen3.5:4b

It fits in the laptop's memory (3.4 GB; the 9.6 GB gemma4:e4b did not), it reads both text and page images, so one model handles OCR and analysis, and Ollama can make it reply in a fixed JSON structure.

2.4 Evaluation dataset

The test set has 45 test cases over 18 real public documents. Each case has one expected field value, one question with its answer, and the page numbers, all checked by hand against the document; none came from the model. 42 cases use original files and 3 use derived files (a 3-page excerpt and one page saved as an image).

Table 3: Coverage of the test dataset

Coverage	Test cases
Languages	English 25, Hindi 7, Kannada 6, Tamil 3, Malayalam 2, English + Punjabi 2
Formats	Searchable PDF 18, scanned PDF 7, mixed searchable and scanned PDF 8, scanned PDF with text layer 3, old-font PDF 3, long 83-page PDF 3, Word 2, image 1
Document types	16 types, including circulars, government orders, scholarship notices, admission brochures, certificates and application forms

No public Telugu government document with real (Unicode) text could be found, so Telugu documents are not tested. Telugu questions and translation are supported and covered by automated tests.

3. Working End-to-End Application

PatraAI is a one-page Streamlit web application. It runs entirely on the user's computer: documents are not sent to any outside service and uploads are not saved to disk.

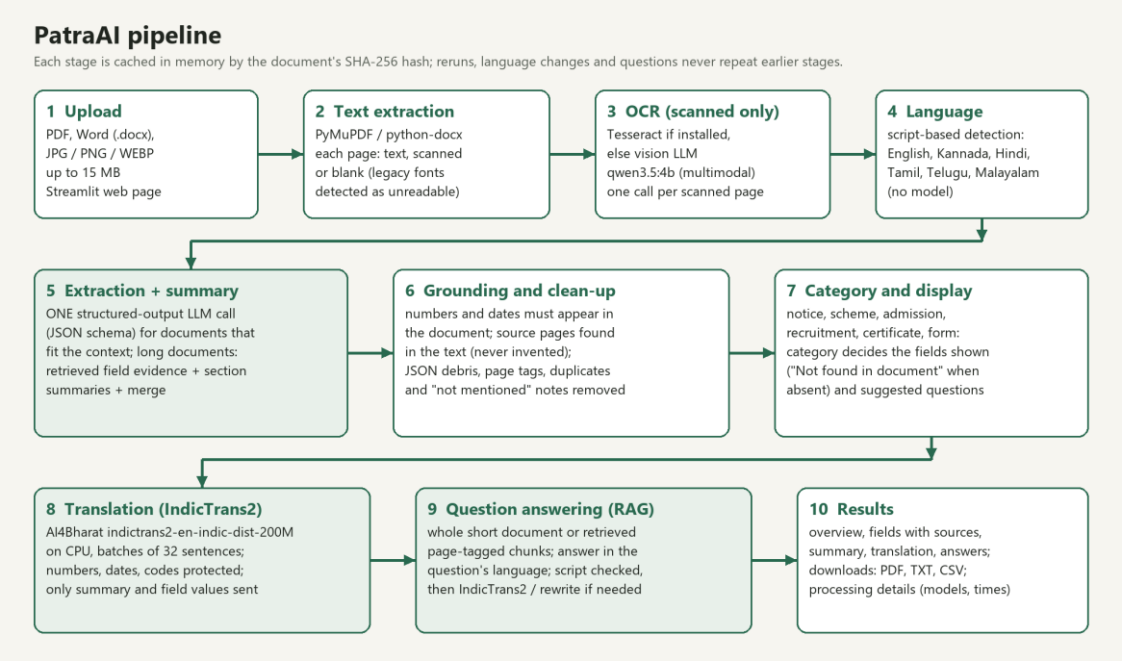


Figure 1**:** The PatraAI pipeline. Shaded steps use AI models; the other steps are ordinary code.

The application reads the text of each page (using OCR only for scanned pages), detects the language, asks the model for the fields and a summary, checks the output against the document, and then translates and answers questions on request. Every step is cached, so changing the language or asking another question does not repeat earlier work.

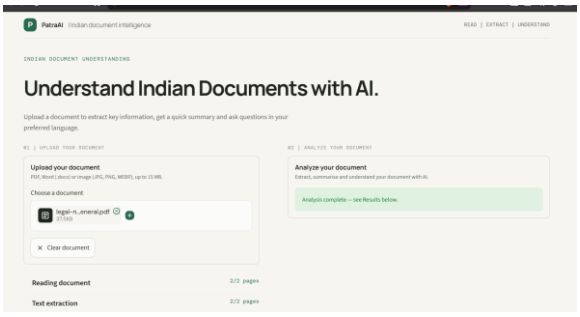


Figure 2**:** Upload panel with page-by-page progress.

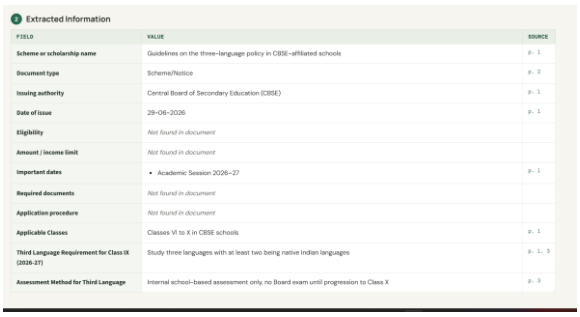


Figure 3**:** Extracted information with source pages.

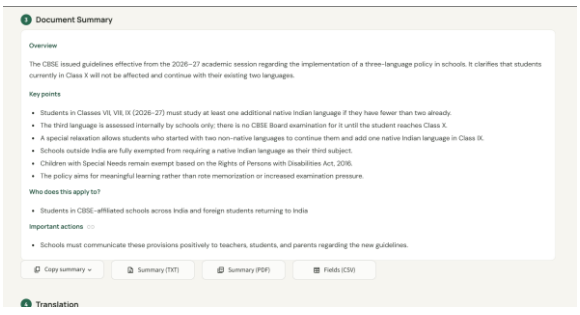


Figure 4**:** Short summary with downloads.

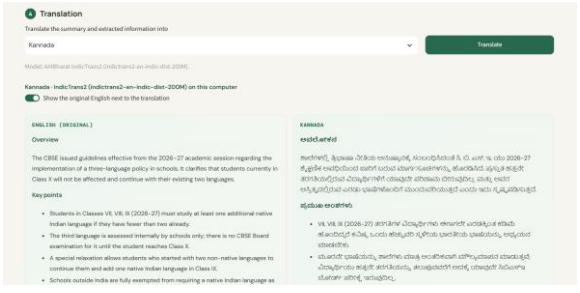


Figure 5**:** Translation with IndicTrans2.

3.1 Features

- **Inputs:** searchable and scanned PDFs of any length, Word (.docx) files and images (JPG, PNG, WEBP), up to 15 MB.
- **Results:** overview, extracted information with source pages, summary, translation, questions and processing details (models used and time taken).
- **Languages:** English, Kannada, Hindi, Tamil, Telugu and Malayalam; each answer can follow the language of its question.
- **Downloads:** all results as a PDF, the summary as TXT or PDF, and the fields as CSV.

4. GenAI, RAG and Agentic Integration

4.1 Structured extraction

For a document that fits in the model's context, one request returns both the fields and the summary as JSON in a fixed structure. For long documents, the passages most likely to contain each field are selected first, and the summary is built from up to five section summaries.

4.2 Question answering with retrieval (RAG)

1. Short documents are given to the model whole.
2. Long documents are split into page-tagged chunks, and the chunks that best match the question are retrieved.
3. The model answers only from those chunks and must cite the pages.
4. If the answer is not there, PatraAI says "Not found in the uploaded document".

4.3 Agentic loop for the answer language

1. Detect the question's language from its script or words.
2. Generate the answer in that language (or the language the user chose).
3. Check the answer's script in code; if it is wrong, translate it with IndicTrans2 or ask the model to rewrite it.
4. If it is still wrong, show an error instead of an answer in the wrong language.

4.4 Checks against hallucination

- A value containing a number that is not in the document is removed.
- A date must be printed in the document; it is never calculated.
- Source pages are found by searching the document text, not taken from the model.

5. Quantitative Testing and Error Analysis

5.1 Method

- Each test document goes through the same pipeline as the website.
- The predicted field and answer are compared with the hand-checked value. Formatting differences are accepted ("₹1,300/-" matches "Rs. 1300"); a different number or date is not.
- "Not found" counts as wrong. Whether the answer cited the right page is scored separately.

The evaluation code is in evaluation/benchmark.py and evaluation/scoring.py. Every case was run once with qwen3.5:4b on the laptop CPU; cases interrupted when the laptop went to sleep were run again. For long Hindi, Kannada and Tamil documents the evaluation searched with the English question only, while the website also searches with a translated question, so those results may understate the website.

5.2 Results

After the failure analysis, two problems were fixed: scanned pages are now read with Tesseract OCR and Indian-language packs (the page's script is detected first, and the vision model is the fallback), and the date check now understands Indian-language month names. All 11 cases on the four documents these fixes affect (two Kannada circulars, a Malayalam gazette and a Hindi circular) were run again; the other 34 cases keep their first-run results. Because the fixes were chosen after looking at the test results, the second figure may be slightly optimistic; a fresh test set would confirm it.

Table 4: Results before and after the fixes

Metric	First run	After fixes
Field-extraction accuracy	64.4% (29 of 45)	75.6% (34 of 45)
QA accuracy	66.7% (30 of 45)	77.8% (35 of 45)
Answer cited the correct page	66.7%	77.8%
Processing errors	0	0

Table 5: Results by document language

Language	Cases	Field accuracy	QA accuracy
English	25	76.0%	88.0%
English + Punjabi	2	100.0%	100.0%
Hindi	7	71.4%	71.4%
Kannada	6	100.0%	83.3%
Malayalam	2	50.0%	50.0%
Tamil	3	33.3%	0.0%

English documents did best. Searchable PDFs reached 72.2% field and 83.3% QA accuracy, while scanned PDFs reached only 85.7% and 85.7%. All Kannada documents needed OCR, and the only Tamil document has 83 pages. Tamil and Malayalam have only 3 and 2 cases, so their figures are rough.

5.3 Failure analysis

All 21 wrong field and answer checks were traced to a root cause by checking the text PatraAI read from the page, whether the right page reached the model, and what the model returned.

Table 6: Root causes of the wrong results

Root cause	Wrong checks	Where it happened
Right page not found in a long document	10	Admission brochure (English); Vacancy circular (Hindi); Public policy document (Tamil)
Model missed a value that was in the text	6	Vacancy circular (Hindi); Government circular (Kannada); Scholarship notification (English); Gazette notification (Malayalam); Application form + certificate format (English); Educational document (English)
Another number taken as the reference number	3	Vacancy circular (Hindi); Government scheme order (English); Gazette notification (Malayalam)
The scorer was too strict	2	Scholarship notification (English); Government scheme order (English)

- **Right page not found in a long document.** Long documents are searched by matching words, so the page with the answer was sometimes not selected. In a 127-page brochure and an 83-page Tamil document the model never saw that page and correctly said it could not find the answer.
- **Model missed a value that was in the text.** The right text reached the model, but it left a value out (for example a list of required documents), said an answer was missing although it was in the text, or a quick text-matching shortcut quoted the wrong amount.

- **Another number taken as the reference number.** Government orders quote many other orders and registration numbers. The model sometimes returned one of these instead of the document's own reference number.
- **The scorer was too strict.** The automatic scorer rejected two correct results because they were written differently from the expected text ("Class I to Class XII" instead of "Classes I to XII", and a correct reference number with another number added). True accuracy is therefore slightly higher than reported.

Wrong page references are not counted above: 10 answers cited the wrong page or none, and 7 of these were also wrong answers.

Table 7: Examples of wrong results

Document type	Check	Expected	Predicted	Cause
Admission brochure (English)	amount	₹ 1300/- | Rs. 1300	(nothing)	Page not found
Public policy document (Tamil)	answer	58,800 | 58800	The document does not contain information about the total number of schools in the Tami...	Page not found
Government scheme order (English)	reference number	G.O.MS.No. 26 | G.O.Ms.No.26	G.O.Rt.No. 326, G.O.Ms.No.9, Comp. No. 2858012	Wrong number picked
Application form + certificate format (English)	required documents	Voter ID | Aadhaar | IT returns | RoR	(nothing)	Model missed it
Government scheme order (English)	answer	Classes I to XII | Class I to XII | I to XII	The "THALLIKI VANDANAM" scheme covers students from Class I to Class XII, including Int...	Too strict

5.4 Speed

The same five documents were timed before and after the speed work. Analysis now takes 23% less time and translation 50% less time, with the same accuracy. The main changes were one combined model call with shorter instructions, reading the prompt in advance while the user looks at the preview, caching every step, and translating in batches.

Table 8: Speed before and after optimisation

Document	Analysis	Translation (Kannada + Hindi)	Fields correct
Government order (1 page)	76 s → 58 s	43 s → 32 s	2/2 → 2/2
Press release (4 pages)	107 s → 84 s	59 s → 36 s	3/3 → 3/3
Word document	92 s → 78 s	121 s → 40 s	1/2 → 1/2
Scheme order (7 pages)	195 s → 148 s	104 s → 62 s	2/3 → 2/3
Hindi circular (7 pages)	149 s → 109 s	53 s → 21 s	2/3 → 2/3
Total	618 s → 476 s	380 s → 191 s	unchanged

5.5 Automated tests

98 automated tests (pytest) check page handling and OCR routing, cleaning of model output, language detection, translation, the scoring rules and the web page itself, using stand-in model services. All 98 pass.

6. Demo, Documentation and Repository

GitHub repository: https://github.com/pavankumarkn04/PatraAI

- **README.md:** setup, models, how the pipeline works, results and limitations.
- **Indian.py and the other .py files:** the web application and its processing code.
- **evaluation/:** the 45 test cases and 18 documents, the scorer, and the results (every check and prediction).
- **tests/ and docs/:** 98 automated tests; this report and the screenshots.

6.1 How to run

1. Install Python 3.11 and Ollama. Accept the IndicTrans2 terms on Hugging Face and put a read token in a .env file (HF_TOKEN=...).
2. Double-click Start PatraAI.bat. It installs missing packages, downloads the models the first time and opens the application in the browser.
3. Double-click Stop PatraAI.bat to stop it.

6.2 Demo script

1. Upload a government PDF and show the page preview.
2. Click Analyze. Show the extracted details with their source pages, and the summary.
3. Change the response language (for example to Kannada); the results appear without analysing again.
4. Ask a question in Kannada or Hindi and show the answer, its language and its source page.
5. Ask something the document does not say; the answer is "Not found in the uploaded document".
6. Upload a scanned page or photo (OCR runs only for scanned pages) and download the results as a PDF.

7. Limitations and Future Work

- **Patram was not run.** It needs a GPU server; comparing it with qwen3.5:4b on the same test cases is the main next step.
- **OCR of poor scans.** Tesseract reads clean scans well, but faint or low-quality scans can still lose words.
- **Long documents.** Word-matching retrieval can miss the right page, especially in Indian-language documents. Three further fixes found in the failure analysis are not yet evaluated: always giving the model page 1, Indian-language keywords for finding fields, and asking evaluation questions through the website's translated search.
- **Evaluation scope.** Only the 11 cases affected by the fixes were run again; Telugu documents are not tested, and Tamil and Malayalam have few cases.
- **Speed.** On a CPU, a typical document takes about 1–2.5 minutes and a 100-page document several minutes; a GPU would make it interactive.

8. Conclusion

PatraAI works end to end on real Indian government documents: it reads searchable, scanned, old-font and Word files, extracts details with their source pages, summarises, translates with IndicTrans2 and answers questions in six languages. On 45 hand-checked test cases it reached 75.6% field-extraction accuracy and 77.8% QA accuracy after fixing the problems found in the failure analysis (first run: 64.4% and 66.7%). Better retrieval in long documents and running Patram on a GPU server are the clearest next steps.

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Appendix A. Test Cases and Results

Field, Answer and Page show whether the expected field, the answer and the cited page were correct. Case numbers follow evaluation/dataset/cases.jsonl in the repository.

Table 9: All test cases and results

No.	Document type	Language	Question	Field	Answer	Page
1	Admission brochure	English	Who issued this information brochure?	Yes	No	No
2	Admission brochure	English	What is the application fee for General Category candidates from Maharashtra State?	No	Yes	Yes
3	Admission brochure	English	What is the parental annual income limit for the reservation scheme for Maharashtra State candidates?	No	Yes	Yes
4	Press release / notice	English	What is the date of the press release?	Yes	Yes	Yes
5	Press release / notice	English	Out of the three languages, how many must be Bhartiya Bhashas?	Yes	Yes	Yes
6	Press release / notice	English	In how many scheduled Bhartiya Bhashas are dedicated R3 textbooks for Class VI being made available?	Yes	Yes	No
7	Vacancy circular	Hindi	Which office issued this circular?	Yes	Yes	Yes
8	Vacancy circular	Hindi	How many posts of Assistant Director (Language) are to be filled?	Yes	No	No
9	Vacancy circular	Hindi	What is the maximum age limit for appointment on deputation?	No	Yes	Yes
10	Vacancy circular	Hindi	Who signed the vacancy circular?	No	No	No
11	Official circular / letter	Hindi	Which department issued this circular?	Yes	Yes	Yes
12	Official circular / letter	Hindi	Which post is to be filled on deputation?	Yes	Yes	Yes
13	Official circular / letter	Hindi	What is the pay matrix range for the post?	Yes	Yes	Yes
14	Government circular	Kannada	Which department issued this circular?	Yes	Yes	Yes
15	Government circular	Kannada	For how many days was the bridge programme conducted in June?	Yes	Yes	Yes
16	Government circular	Kannada	On which Saturday of every month should the head teachers' meeting on FLN be held?	Yes	Yes	Yes
17	Government circular	Kannada	What is the date of this circular?	Yes	No	Yes
18	Government circular	Kannada	How many self-learning activity modules were prepared for students?	Yes	Yes	Yes
19	Government circular	Kannada	Which module is scheduled for January 2027?	Yes	Yes	Yes
20	Scholarship notification	English	What is the parental annual income limit for this scholarship?	No	Yes	Yes
21	Scholarship notification	English	What is the maximum scholarship amount per annum?	Yes	No	No
22	Scholarship notification	English	To whom should applications of Backward Classes (BC) students be sent?	Yes	Yes	No
23	Scholarship notification	English	What is the letter number?	Yes	Yes	Yes
24	Government scheme order	English	How much financial assistance is given per child per annum under the scheme?	No	Yes	Yes
25	Government scheme order	English	How much is deducted at source per student?	Yes	Yes	Yes
26	Government scheme order	English	Which classes are covered by the scheme?	Yes	No	Yes
27	Public policy document	Tamil	How many schools did the School Education Department have as of March 2024?	No	No	No
28	Public policy document	Tamil	How many students were there as of March 2024?	Yes	No	No

No.	Document type	Language	Question	Field	Answer	Page
29	Public policy document	Tamil	What share of total state expenditure is allocated to school education?	No	No	No
30	Gazette notification	Malayalam	What is the G.O. number of this notification?	No	Yes	Yes
31	Gazette notification	Malayalam	Under which Act is this notification issued?	Yes	No	Yes
32	Notification with schedule table	English	What is the last date for online application submission and fee payment?	Yes	Yes	Yes
33	Notification with schedule table	English	When is the certificate verification of eligible candidates?	Yes	Yes	Yes
34	Government order	English	What amount is administratively sanctioned in this order?	Yes	Yes	Yes
35	Government order	English	Who signed this government order?	Yes	Yes	Yes
36	Official proceedings / letter	English	What honorarium does the District Collector receive as chairperson of the district committee?	Yes	Yes	Yes
37	Official proceedings / letter	English	On which dates were certificate verification and interviews originally scheduled for all institutions?	Yes	Yes	Yes
38	Government certificate (prescribed format)	English	Who can issue this certificate?	Yes	Yes	Yes
39	Government certificate (prescribed format)	English	Which website gives the latest guidelines on the Central List of OBCs?	Yes	Yes	Yes
40	Application form + certificate format	English	What is the gross annual family income limit for the EWS certificate?	Yes	Yes	Yes
41	Application form + certificate format	English	Above what size of agricultural land is a family excluded?	No	Yes	No
42	Application form	English + Punjabi	What is the RTS service number of this form?	Yes	Yes	Yes
43	Application form	English + Punjabi	When was this form last revised?	Yes	Yes	Yes
44	Educational document	English	Which BharatGen model is suggested for the Indian Document Understanding Tool?	Yes	Yes	Yes
45	Educational document	English	How many representative test cases are required?	No	Yes	Yes